

Group Reflection Journal

ITAI 1374 | NeuralCrew | Telco Customer Churn Midterm

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Individual reflections compiled from the team members' submitted statements.

Individual Reflection — Patrick Ray

Working on our Telco Customer Churn Midterm Project helped me better understand how important data preparation is before building a machine learning model. I learned that even when a dataset contains useful information, it still needs to be cleaned, organized, and properly processed before it can be used effectively.

One of the most important things I learned was why we need to separate the dataset into 70% training data and 30% testing data. I now understand that exploratory data analysis and preprocessing decisions should be based on the training data while keeping the testing data untouched. This helps prevent data leakage and allows us to evaluate our future machine learning models more fairly.

I also gained a better understanding of handling missing values, converting categorical information into numerical values, scaling features, and balancing an imbalanced dataset. Running and reviewing the notebook in Google Colab helped me see how these different preprocessing techniques work together to prepare the data for machine learning.

One of my biggest challenges during this project was coordinating everyone's contributions while making sure we followed the professor's instructions and deadlines. As the group leader, I was responsible for communicating with team members, collecting their completed work, and organizing the files for our final submission. I learned that good communication and organization are just as important as understanding the technical work, especially when group members have different schedules and responsibilities.

Overall, this project helped me become more comfortable working with Jupyter notebooks and understanding the machine learning workflow. It also strengthened my confidence in organizing a group project and bringing everyone's contributions together. I look forward to using our prepared dataset in the final project, where we will begin training and evaluating machine learning models.

Reflection from Approved Statement — Viviana Zamora

Viviana confirmed that the group may use her previously submitted information. The following is her existing learning statement, reproduced without adding new personal experiences:

“Through this work, I learned why data cleaning should be done in code and why testing data must remain separate from training-data analysis.”

Her submitted statement also documents her role in selecting the dataset, contributing to Python coding and data cleaning, and revising project documents. This paragraph describes her contribution, not an additional personal reflection.

Individual Reflection — Dustin Mastrota

Reviewing our Telco Customer Churn project helped me appreciate how much work goes into preparing a dataset before it can be used for machine learning. I found the entire process interesting, especially seeing how the notebook connected the preprocessing steps into one workflow. Every week in this course continues to surprise me as we work with different Jupyter notebooks and see the outcomes. This project gave me another opportunity to connect the code with its purpose instead of only looking at the final results.

One of my biggest areas of growth is recognizing and understanding the coding involved. I am getting more comfortable following what the code is doing, but just when I feel like I am getting the hang of it, I come across a technique or function I have not seen before. That can be challenging, but it is also exciting because it shows me how much more there is to learn. During this review, I focused on understanding why each preprocessing step was used so I could explain it clearly in my own words.

The project also helped me understand that preparing data is not just about making it look cleaner. The choices made during preprocessing affect how the information can be used later. Reviewing the handling of missing values, categorical encoding, scaling, and class balancing helped me see that each step has a specific purpose. I also developed a better understanding of why the testing data needs to remain separate and why completing preprocessing does not automatically mean a model will make accurate predictions.

Another meaningful part of this project was seeing what my teammates accomplished. I am proud of the work they produced and how smoothly our individual contributions came together. Staying in contact with the team helped me understand their roles and write a review that reflected the overall project. Although my role focused on analysis rather than developing the original code, it gave me valuable practice interpreting technical work and explaining why it matters. I finished this portion feeling more confident in my understanding and excited to see how our prepared dataset will be used in the final project.

Individual Reflection — Funsho Orogun

I learnt that this dataset can be used to predict customer churn using a binary classification model. I would split the data into training and testing sets, train models such as Logistic Regression or Random Forest, and evaluate their performance using metrics like accuracy, precision, recall, and F1-score. I also learnt the importance of tuning models and interpreting the results so businesses can identify customers who are likely to leave and take action to improve retention.